

CUSTOMER ANALYTICS

Customer Shopping Behavior Analysis

Insights from 3,900 transactions to guide subscription, discount, and marketing strategy

Project Overview

Why we undertook this analysis

We analyzed shopping behavior across 3,900 customer transactions to uncover patterns in spending, loyalty, and product preference — and translate them into concrete actions for subscriptions, discounting, and marketing spend.

- Understand where revenue concentrates today
- Identify who our most loyal customers are
- Evaluate whether discounting is helping or hurting margin
- Surface the levers stakeholders can pull this quarter

SCOPE OF DATA

3,900

customer transactions

18

data fields analyzed

4

product categories

10

business questions answered

How We Got Here

From raw transactions to stakeholder-ready insight

01

Clean & Prepare

Cleaned data in Python: imputed missing ratings, standardized fields, engineered age and frequency groups.

02

Query & Analyze

Loaded data into PostgreSQL and ran 10 targeted business queries covering revenue, loyalty, and discounting.

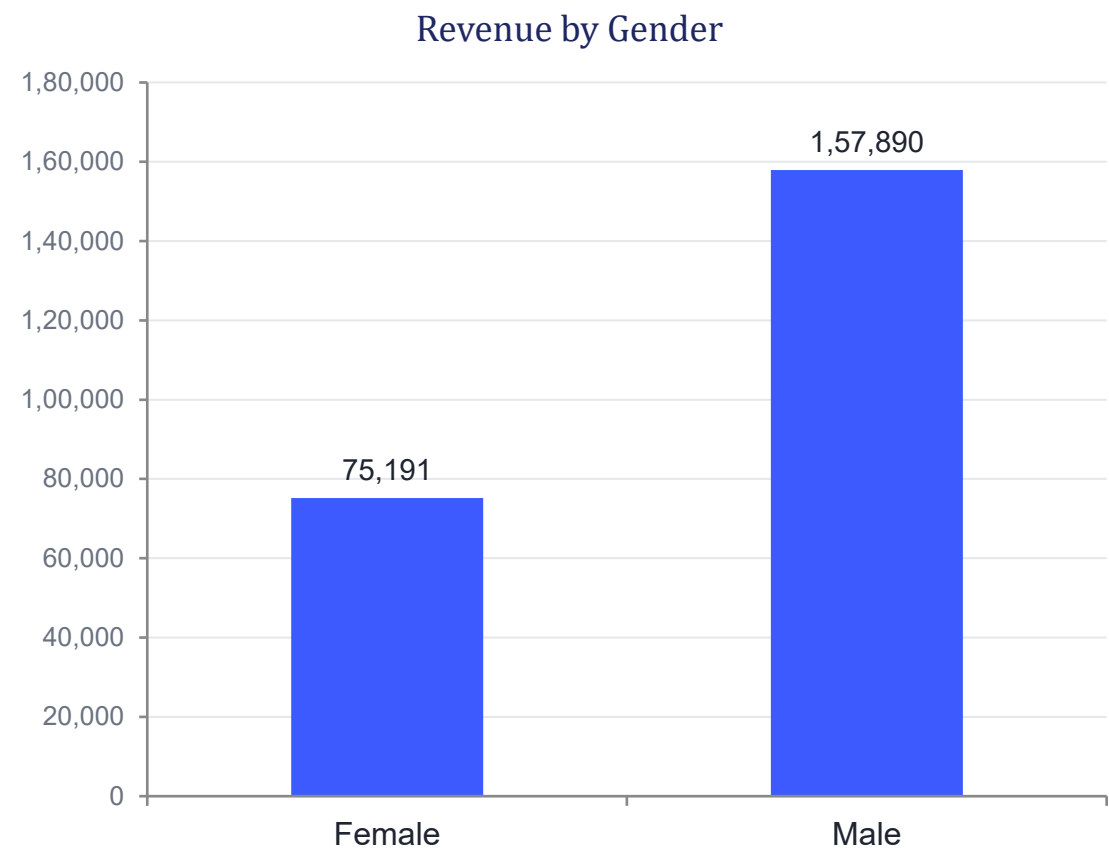
03

Visualize & Present

Built an interactive Power BI dashboard so stakeholders can filter by segment, category, and shipping type.

Where Revenue Comes From

Gender split and category mix



Male customers generate

2.1x

more revenue than female customers (\$157,890 vs. \$75,191) — worth investigating whether this reflects basket size, category mix, or acquisition channel.

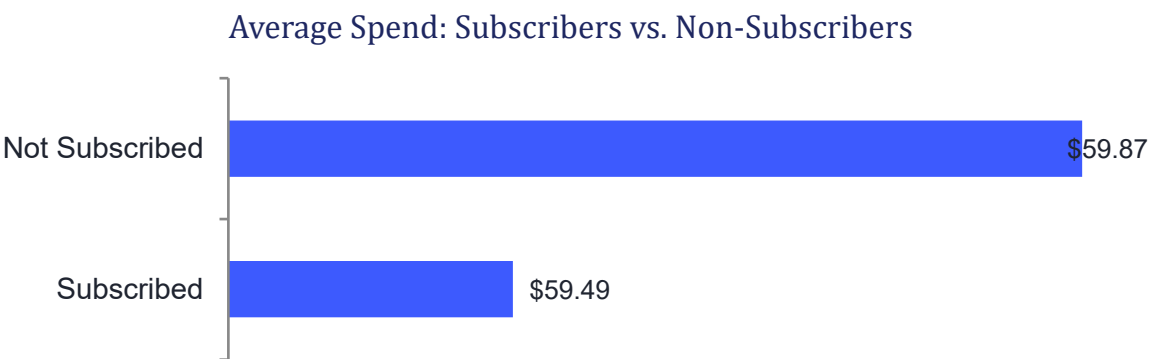
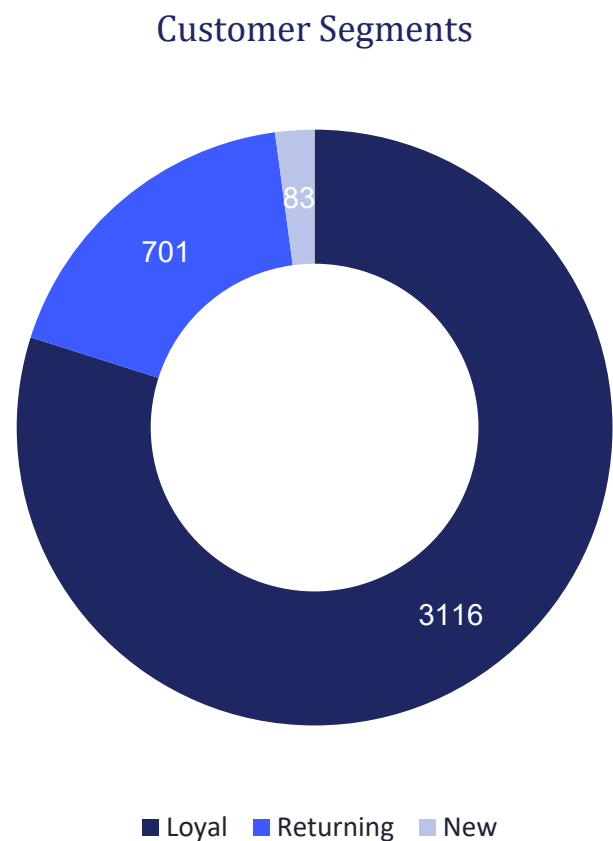
Top-selling category

Clothing

Blouses and pants lead orders (171 each), followed closely by shirts (169) — Clothing is the anchor category across the store.

Loyalty & Subscriptions

Most customers are loyal — but few are subscribed

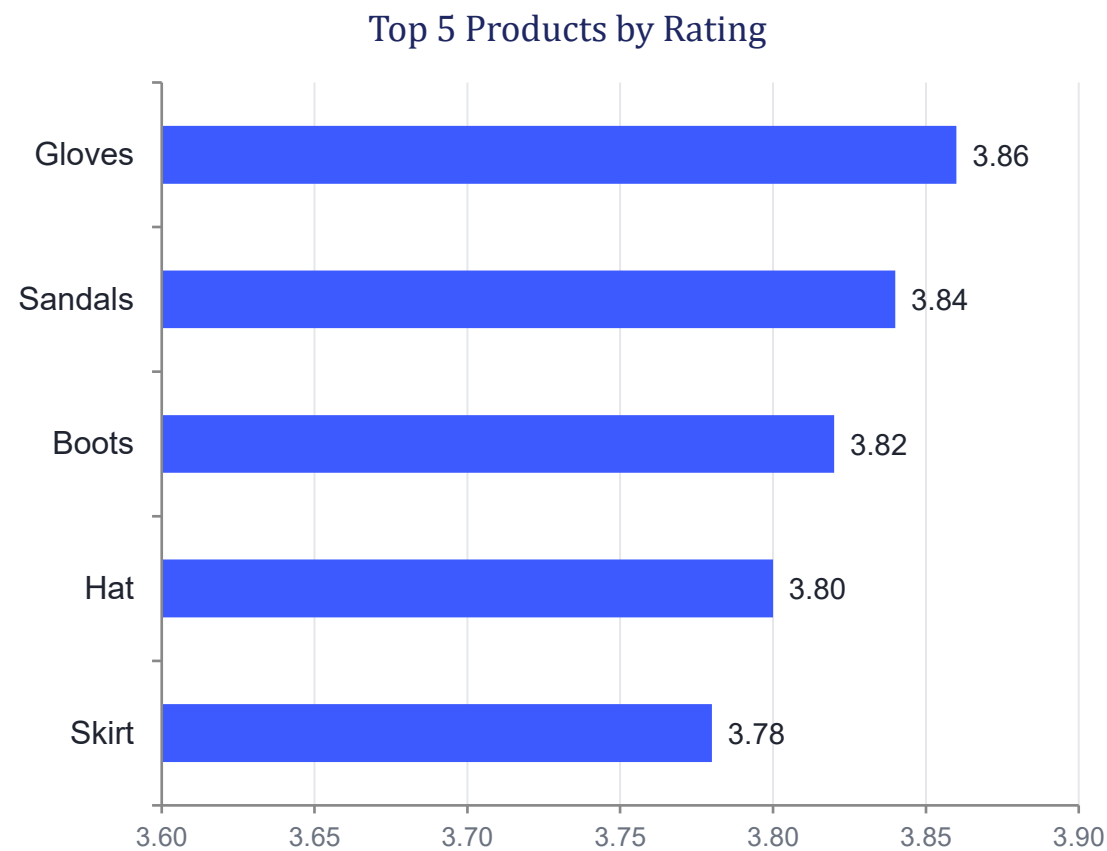


The subscription gap

Only 1,053 of 3,900 customers (27%) are subscribed, yet subscribers spend almost the same per order as non-subscribers (\$59.49 vs. \$59.87). Non-subscribers still generate 73% of total revenue (\$170,436) — the biggest untapped opportunity is converting loyal, non-subscribed customers.

Product Performance

What customers rate highest and buy most



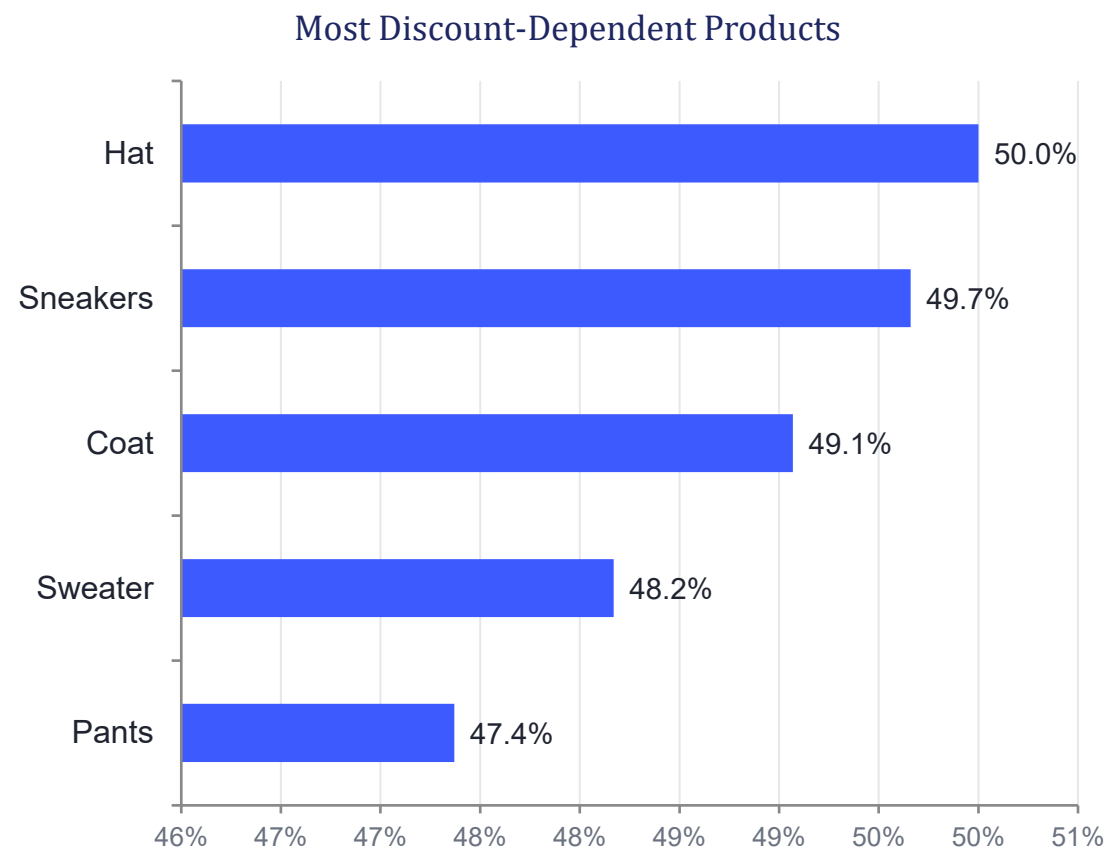
Category leaders by orders

Clothing	Blouse — 171 orders
Accessories	Jewelry — 171 orders
Outerwear	Jacket — 163 orders
Footwear	Sandals — 160 orders

Ratings cluster tightly (3.78–3.86) — no product is underperforming badly, but footwear and accessories are the categories to feature first in campaigns given both strong ratings and volume.

Discounting & Shipping Behavior

Where discount reliance is highest



839

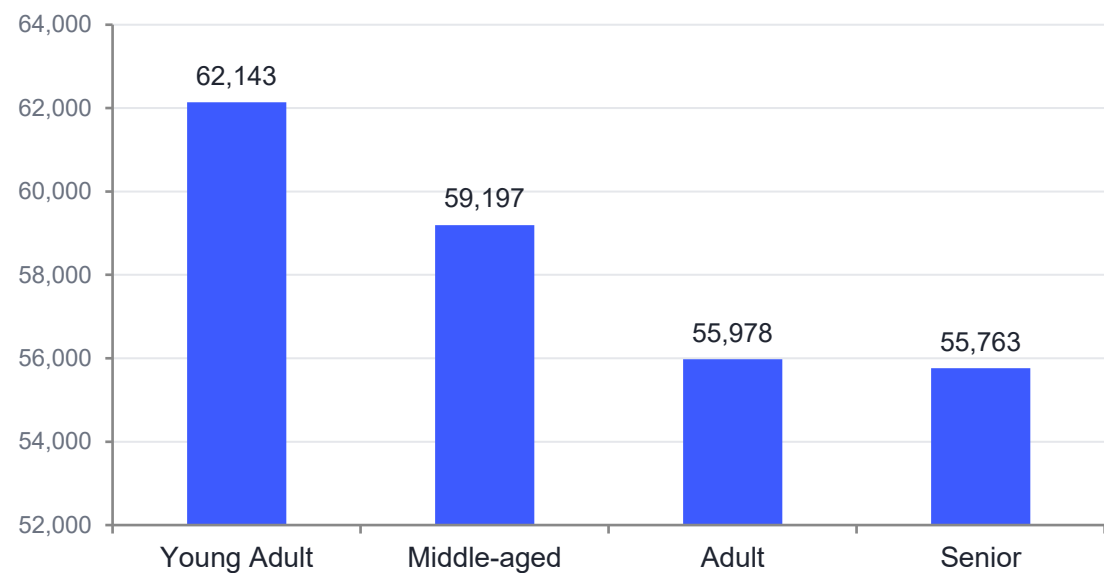
customers used a discount and still spent above the \$59.76 average order — discounting isn't only attracting bargain hunters.

Shipping type barely moves spend

Express shipping orders average \$60.48 vs. \$58.46 for Standard — a \$2 gap. Customers are not paying a meaningful premium to ship faster, so express fees are unlikely to offset added logistics cost.

Revenue by Age Group

Spend is broad-based across generations



Revenue is remarkably even across age groups (within an 11% band) — age-targeted campaigns should focus on message and channel, not on chasing one dominant demographic.



Recommendations

Five actions to act on this analysis

01

Boost Subscriptions

Promote exclusive benefits for subscribers — the segment already spends at parity but is only 27% of customers.

02

Build Loyalty Programs

Reward repeat buyers to move Returning and New customers into the Loyal segment.

03

Review Discount Policy

Balance sales boosts with margin control on the most discount-dependent products.

04

Sharpen Product Positioning

Feature top-rated, high-volume products — footwear and accessories — in campaigns.

05

Target by Category, Not Age

Focus marketing spend on high-revenue categories and express-shipping users rather than a single age band.

Thank You

Questions & discussion